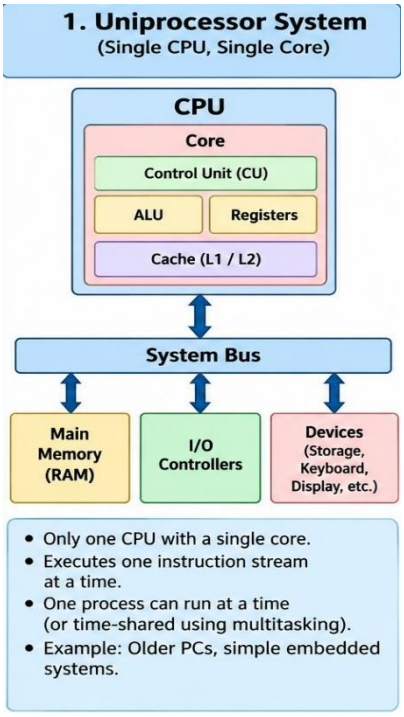
Uniprocessor System

1. Introduction

A **uniprocessor system** is a type of computer system that has a **single central processing unit (CPU)**. Which executes only **one instruction at a time**. All tasks, including user applications and system processes, are handled by this **single processor**.

Uniprocessor systems are commonly found in **personal computers, mobile devices, and embedded systems**. These type of system is used for simple user application software like **web browsing, word processing, and basic gaming etc**.

Example : Intel 80386, 80486, Pentium I, II, III



3. Advantages of Uniprocessor Systems

- **Cost-Effective**
  - Uniprocessor systems are **cheaper** compared to multiprocessor setups.
  - They are ideal for applications that do not require high-performance computing.
- **Lower Power Consumption**
  - Since only one CPU is used, power consumption is relatively **lower** compared to multiprocessor systems.

Pushkar Raj
PR
Pushkar Raj

- A multicore CPU consume less power than single core processor
- Suitable for **mobile devices and battery-powered systems**.

➢ **Simpler Design & Maintenance**

- The **hardware and software** complexity is lower, making **design, implementation, and maintenance** easier.
- Efficient for applications that require **single-threaded and multi-threaded processing**.

4. Disadvantages of Uniprocessor Systems

➢ **Limited Performance & Scalability**

- A **single CPU can process only one instruction at a time**, limiting performance.
- As more tasks are added, **context switching increases**, slowing down execution.
- Not suitable for **high-performance computing applications** like simulations, AI, or large-scale databases.

To improve processing power or speed of uniprocessor, multicore processor technology is introduced in 2005 by Intel and AMD.

Multicore uniprocessor system : A single processor with many separate processing unit inside it. A multicore processor may have two core (dual core), four core (quad core) etc. which enhance processing power many times.

Every core of this processor have separate control unit, ALU, set of registers and all task is divided into subtask (called thread) is executed on different core and all result are joined together to form a final result.

Example:

Intel I3 : 2 core (till 7th gen), 4 core.
Intel I5 : 4 core (till 7th gen) 6 core.
Intel i7 : 4 core (till 7th gen) , 8 core

Advantages of multicore Uniprocessor Systems:

- Fast context switching result high performance multitasking.
- Complex and high processing need software run smoothly.

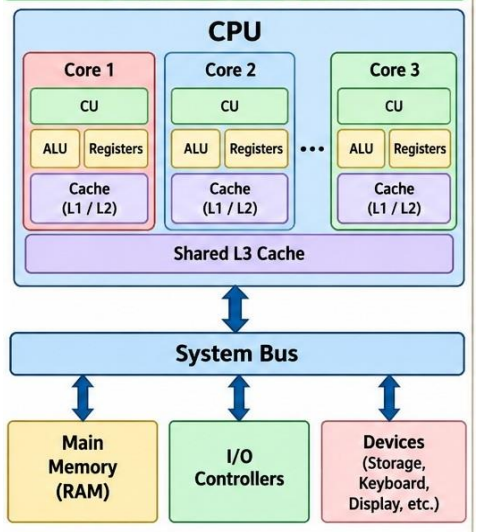


- Mostly gaming computer/laptops are multicore system.
- Power consumption reduced drastically.

Pushkar Raj
PR
Pushkar Raj

2. Uniprocessor System with Multiple Cores

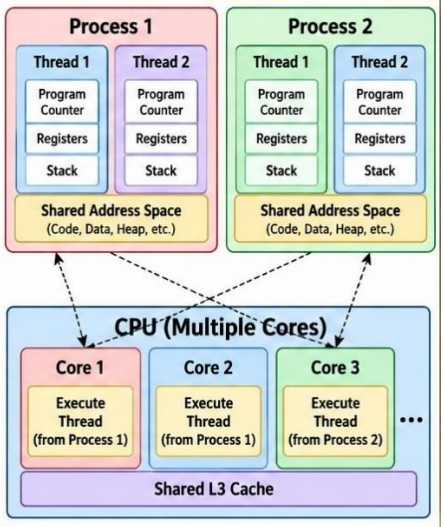
(Single Chip, Multiple Cores)



- One physical CPU chip with multiple cores.
- Each core can execute its own instruction stream.
- Multiple processes/threads can run in parallel on different cores.
- Cores share common resources like L3 cache, memory controller, and I/O.
- Example: Modern CPUs (Intel Core i5/i7/i9, AMD Ryzen).

3. Processes and Threads on Uniprocessor with Multiple Cores

(Execution View)



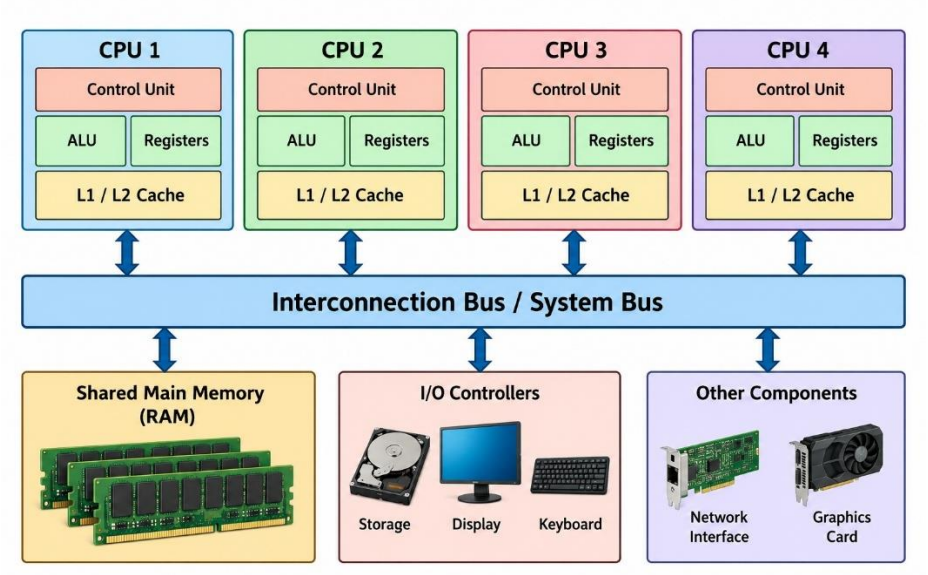
- A process is a program in execution with its own address space (code, data, heap, resources).
- A thread is a lightweight unit of execution within a process.
- All threads of a process share the same address space, but each thread has its own program counter, registers, and stack.
- On a multi-core CPU, threads from the same or different processes can run in parallel on different cores.
- The operating system schedules threads to available cores.

Multiprocessor System:

A **multiprocessor system** is a type of computer system that includes two or more processors (CPUs) sharing the same memory and operating under a single operating system. These systems are designed to enhance computing power, improve performance, and ensure reliability by allowing multiple processors to work together.

Characteristics of Multiprocessor Systems

1. **Multiple CPUs** – These systems have two or more processors working simultaneously.
2. **Shared Memory** – All processors share a common main memory, enabling faster communication.
3. **Single OS Control** – The entire system operates under one operating system that manages resources and tasks.
4. **Parallel Processing** – Tasks can be divided among processors, improving performance and efficiency.
5. **Increased Reliability** – Even if one processor fails, others can continue working, reducing downtime.



Types of Multiprocessor Systems

1. Symmetric Multiprocessing (SMP)

- In this model, all processors share memory and are treated equally by the OS.
- Each processor runs tasks independently but can communicate with others.
- Example: **Modern multicore processors like Intel Core i7 and AMD Ryzen.**

2. Asymmetric Multiprocessing (AMP)

Pushkar Raj
PR
Pushkar Raj

- Here, one processor is designated as the master, controlling the system and assigning tasks to other processors.
- The subordinate processors execute tasks but do not manage the system.
- Example: **Older mainframes and some embedded systems use AMP architecture.**

Advantages of Multiprocessor Systems

- **Higher Performance** – Tasks can be divided among multiple processors, reducing execution time.
- **Better Reliability** – Failure of one processor does not cause system failure, ensuring continued operation.
- **Efficient Resource Utilization** – Processors share memory and resources, preventing bottlenecks.
- **Improved Parallel Processing** – Useful for applications requiring complex computations, such as AI, ML, and simulations.

Disadvantages of Multiprocessor Systems

- **Complexity in Design** – Managing multiple processors and memory access requires a sophisticated OS.
- **High Cost** – More processors mean higher hardware costs.
- **Synchronization Issues** – Processors must coordinate tasks properly to avoid conflicts.

Real-Life Examples of Multiprocessor Systems

1. **Supercomputers** – Used for scientific research, weather forecasting, and AI computations.
2. **Servers** – Data canters use multiprocessor servers for handling multiple user requests simultaneously.
3. **Medical Imaging Systems** – CT scans and MRI machines rely on multiprocessor architectures for faster image processing.

Pushkar Raj
PR
Pushkar Raj

1. Off-line Data Entry

Off-line data entry is a method of entering data into a computer system where the data is collected and stored first, without being directly connected to the main computer system. The stored data is transferred to the main system later for processing.

Example: A data entry operator enters students' examination details into a file and submits the file to the examination department for processing later.

Key point: Data is entered first and processed later.

2. On-line Processing

On-line processing is a method in which a computer processes data through a connected system, allowing users to access updated information while working. The data is processed as users submit their requests, although the response may not always be instantaneous.

Example: When a student updates their profile on a college website, the information is stored in the database and can be accessed through the online system.

Key point: Data is processed through a connected computer system, and updated information is made available online.

3. Online Data Entry

Online data entry means entering data directly into a computer system through a connected device or application. The entered data is usually transmitted to a central server or database.

Example: A student fills out an online admission form and submits personal details through a college website.

Key point: Data is entered directly into an online system.

4. Real-time Processing

Real-time processing is a method in which data is processed immediately or within a specified time limit so that the system can respond to events without unacceptable delay.

Example: Airbags in vehicle, missile targeting etc

5. Time-sharing Processing

Time-sharing processing allows multiple users to use the same computer system seemingly at the same time. The operating system gives each user or task a small amount of CPU time in turn, switching rapidly between them.

Example: Several students use different terminals to access the same central computer in a computer laboratory.

Pushkar Raj
PR
Pushkar Raj

Key point: CPU time is shared among multiple users or tasks.

6. Electronic Mail (E-mail)

Electronic mail, commonly called e-mail, is a method of sending and receiving digital messages through the internet or another computer network. Users can send text, documents, images and other files using an e-mail service.

Example: A teacher sends lecture notes and an assignment to students using Gmail.

Key point: E-mail enables the electronic exchange of messages and files.

Important fields of an e-mail:

- To: Contains the e-mail address of the main recipient.
- From: Shows the e-mail address of the sender.
- CC (Carbon Copy): Sends a copy of the message to additional recipients. Their addresses are generally visible to other recipients.
- BCC (Blind Carbon Copy): Sends a copy to additional recipients without revealing their addresses to other recipients.
- Subject: A short heading describing the purpose of the message.
- Message Body: The main text written by the sender.
- Attachment: A file, such as a PDF, image, Word document or spreadsheet, sent along with the e-mail. The recipient can open or download the attached file.

History of e-mail: In 1965, early computer mail programs allowed users to exchange messages on the same computer system. In 1971, American computer engineer Ray Tomlinson developed networked e-mail over ARPANET and introduced the use of the @ symbol in e-mail addresses. In India, e-mail was used in academic and research networks before the public commercial internet became available. On 15 August 1995, Videsh Sanchar Nigam Limited (VSNL) launched commercial internet services in India, helping more people access e-mail. Gmail, introduced by Google in 2004, later made web-based e-mail widely popular.

Advantages of e-mail:

1. It provides fast and low-cost communication.
2. It allows files and documents to be shared easily.
3. A single message can be sent to multiple recipients.
4. Messages can be stored and retrieved when required.
5. It enables communication across the world

7. Teletext

Teletext is an information service that displays text-based information on a television screen. Traditionally, it provides information such as news headlines, weather reports, sports results and television schedules through dedicated pages.

Pushkar Raj
PR
Pushkar Raj

Example: A viewer selects a teletext page on a compatible television to read weather information.
Key point: Teletext provides text-based information through a television system.

8. Teleconferencing

Teleconferencing is a method of conducting meetings between people at different locations using communication technology. It may involve audio, video or both, allowing participants to communicate without being physically present in the same place.
Example: A college conducts an online faculty meeting using Zoom or Google Meet.
Key point: Teleconferencing enables people in different locations to hold meetings remotely.

Quick Revision Table

Topic	Main purpose
Off-line data entry	Enter data for later processing
On-line processing	Process data through a connected system
Online data entry	Enter data directly into an online system
Real-time processing	Respond to data within a required time limit
Time-sharing processing	Share CPU time among multiple users or tasks
Electronic mail	Exchange digital messages and files
Teletext	Display text-based information on television
Teleconferencing	Conduct remote meetings using communication technology

Remember the difference: Online data entry describes how data is entered, whereas on-line processing describes how data is processed through a connected system.